Sri Chaitanya IIT Academy.,India.

A.P ✪ T.S ✪ KARNATAKA ✪ TAMILNADU ✪ MAHARASTRA ✪ DELHI ✪ RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr.Super60_NUCLEUS & STERLING_BT	**Paper -1(Adv-2021-P1-Model)**	**Date: 28-07-2024**
Time: 09.00Am to 12.00Pm	**RPTA-03**	**Max. Marks: 180**

28-07-2024_Sr.Super60_NUCLEUS&STERLING_BT_Jee-Adv(2021-P1)_RPTA-03_Syllabus

PHYSICS : Ray Optics: Refraction at plane and spherical surfaces; Geometrical Optics: Total internal reflection; Thin lenses; Combinations of mirrors and thin lenses; Magnification. Experiments: focal length of a convex lens and convex and concave mirrors using UV method (parallax method), The plot of the angle of deviation vs angle of incidence for a triangular prism. Refractive index of a glass slab using a travelling microscope. Deviation and dispersion of light by a prism (Important for ADVANCED)

CHEMISTRY : Alkene & Alkyne: Preparation, properties and reactions of alkenes and alkynes.

Physical properties of alkenes and alkynes (boiling point, density and dipole moments); Acidity of alkynes; Acid catalysed hydration of alkenes and alkynes (excluding the stereochemistry of addition and elimination); Reactions of alkenes; Preparation of alkenes and alkynes by elimination reactions; Electrophilic addition reactions of alkenes with X2, HX, HOX (X=halogen); Effect of peroxide on addition reactions; cyclic polymerization reaction of alkynes ,Addition reactions of alkynes; Metal acetylides. Reactions of alkenes with KMnO4 and ozone; Reduction of alkenes and alkynes

MATHEMATICS : AOD

Name of the Student: ____________________ **H.T. NO:** | | | | | | | |

JEE-ADVANCED-2021-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** Max Marks: 180

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 10)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 11 – 16)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 17 – 19)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 20 – 23)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 24 – 29)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 30 – 35)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 36– 38)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 39 – 42)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 43 – 48)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 49 – 54)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 55 – 57)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

PHYSICS **Max Marks: 60**

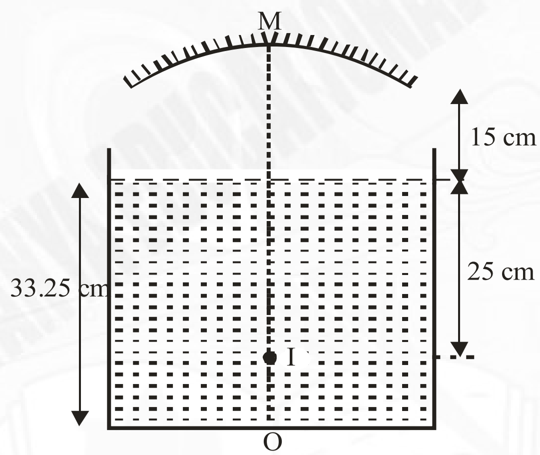
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)

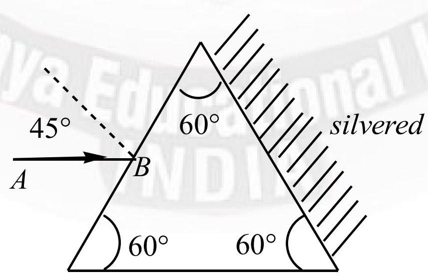
- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
- Full Marks: **+3** (If the answer is correct)
- Zero Marks: **0** (If the question is unanswered)
- Negative Marks: -1 (In all other cases)

1. The adjoining fig. shows object O, final image I formed after two refractions and one reflection is also shown in fig. Then the focal length of the mirror (in cm) is (μ of liquid is $\frac{4}{3}$).



A)10 **B)**15 **C)**18 **D)**25

2. A equilateral prism is made of a transparent material of refractive index $\sqrt{2}$. A ray of light AB is incident at 45^0 as shown. The net deviation in the path of ray when it comes out of prism is



A) 135^0 **B)**120^0 **C)** 30^0 **D)** 150^0

3. A ball is dropped from a height of 20 m above the surface of water in a lake. The refractive index of water is 4/3. A fish inside the lake, in the line of fall of the ball, is looking at the ball. At an instant, when the ball is 18.2m above the water surface, the fish sees the speed of ball as [Take $g = 10 m/s^2$]

A) 10 m/s **B)** 12 m/s **C)** 8 m/s **D)** 21.33 m/s

4. A double convex lens forms a real image of an object on a screen which is fixed. Now the lens is given a constant velocity 0.5 m/s along its axis and away from the screen. For the purpose of forming a sharp image always on the screen, the object is also required to be given an appropriate velocity. The velocity of the object at the instant the size of the image is half the size of the object.

A) 1 m/s **B)** 2 m/s **C)** 1.5 m/s **D)** 4 m/s

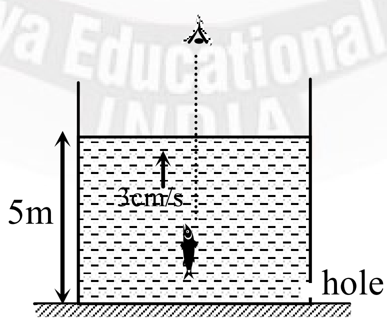
SECTION-II
(PARAGRAPH WITH NUMERICAL VALUE TYPE)

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks:* +2** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

Question Stem for Question Nos. 5 and 6

Question Stem

A tank having cross sectional area A has a hole at the bottom of area of cross section $A_1 = A/1000$. Bottom of the tank is a plane mirror. The tank contains water of refractive index $\frac{4}{3}$. At the instant, when height of the water in the tank is 5m, a fish is rising vertically in the tank with a velocity 3cm/sec toward the surface. (Assume steady flow) $\left(g = 10 m/s^2\right)$



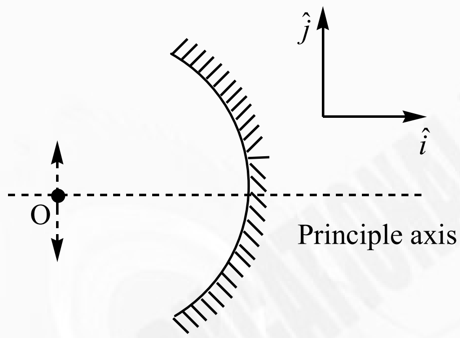
5. The velocity with which surface is falling down inside container is in cm/s.

6. The velocity of the fish as observed by the observer looking directly at the fish is in cm/s.

Question Stem for Question Nos. 7 and 8

Question Stem

A point object is placed at a distance 5R/3 from the pole of a concave mirror. R is the radius of curvature of mirror. Point object oscillates with amplitude of 1 mm perpendicular to the principle axis.



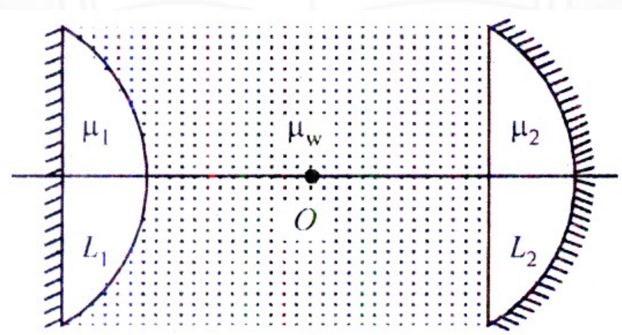
7. The amplitude of image is in mm.

8. position of image when object is at O is –xR, find X.

Question Stem for Question Nos. 9 and 10

Question Stem

A cylinder tube filled with water $\left(\mu_w = \frac{4}{3}\right)$ is closed at its both ends by two, silvered plano convex lenses as shown in the figure. Refractive index of lenses L_1 and L_2 are $\mu_1 = 2.0$ and $\mu_2 = 1.5$ while their radii of curvature are $R_1 = 5\,cm$ and $R_2 = 9\,cm$ respectively. A point object is placed somewhere at a point O on the axis of cylindrical tube. It is found that the object and image coincide each other.



9. The position of object W.r.t lens L_1 is in cm.

10. The position of object W.r.t lens L_2 is in cm.

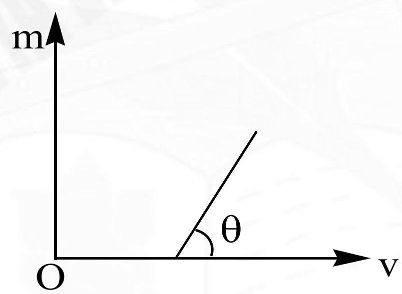
SECTION-III
(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks: +3*** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks: +2*** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks: +1*** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks: 0*** If unanswered;
- ***Negative Marks: -2*** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;
Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and

Choosing any other option(s) will get -2 marks.

11. Figure shows variation of modulus of magnification m(produced by a thin convex lens) and distance v of image from pole of the lens. Which of the following statements are correct?



A) Focal length of the lens is equal to intercept on v-axis.

B) Focal length of the lens is equal to Inverse of slope of the line

C) Magnitude of intercept on m-axis is equal to unity

D) None of above

12. A thin biconvex lens is prepared from glass of refractive index $\mu_2 = \frac{3}{2}$. The two convex surfaces have equal radii of 20 cm each. One of the surfaces is silvered from outside to make it reflecting. The lens now is placed in a large medium of refractive index $\mu_1 = 5/3$. It acts as a

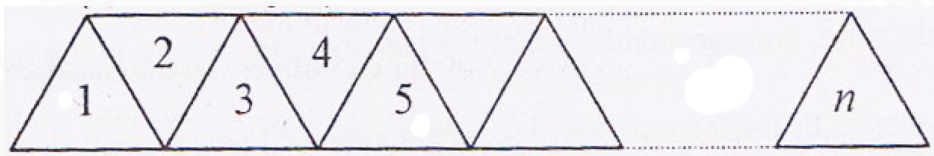
A) Converging mirror

B) Diverging mirror

C) Concave mirror of focal length 12.5 cm

D) Convex mirror of focal length 12.5cm

13. n identical equilateral prisms are kept in contact as shown figure. If deviation through a single prism is δ. Then (n, m are integers)



A) If $n = 2m$, the net deviation through system of n prisms is zero

B) If $n = 2m+1$, the net deviation through system of n prisms is δ

C) If $n = 2m$, the net deviation through system of n prisms is δ

D) If $n = 2m+1$, the net deviation through system of n prisms is zero

14. In the figure light is incident at an angle θ which is slightly greater than the critical angle. Now, keeping the incident fixed a parallel slab of refractive index n_3 is paced on surface AB. Which of the following statements are correct

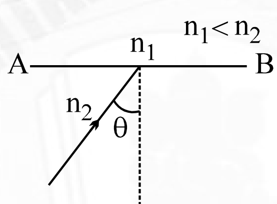


Fig. (a)

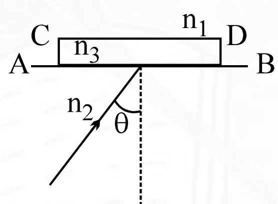


Fig. (b)

A) total internal reflection occurs at AB for $n_3 < n_1$

B) total internal reflection occurs at AB for $n_3 > n_1$

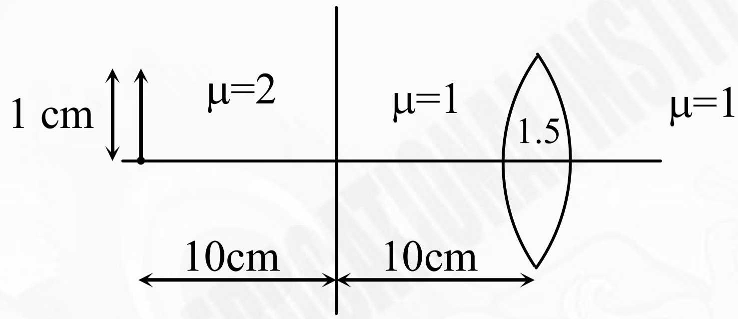
C) the ray will return back to the same medium for all values of n_3

(D) total internal reflection occurs at CD for $n_3 < n_1$

15. The distance between a screen and an object is 120 cm. A convex lens is placed closed to the object and is moved along the line joining object and screen, towards the screen. Two sharp images of the object are found on the screen. The ratio of magnification of two real images is 1:9. Then,

A) focal length of the lens is 22.5 cm
B) smaller image is brighter than the larger one
C) larger image is brighter than the smaller one
D) brightness of both the images is same

16. An object of length 1 cm is placed on a principle axis of bi-convex lens of radius 5 cm. Distance between the lens and object is 20 cm. space between the lens and object is filled with medium of two different refractive index2 and 1 as shown in the figure. Boundary of both medium is mid-way between the object and lens as shown in figure. (consider all rays are paraxial)

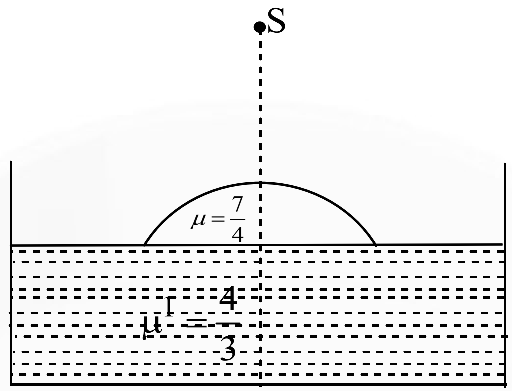


A) The image will be formed at distance of 7.5 cm from the optical centre.
B) The image will be formed at distance of 10 cm from the optical centre.
C) The size of the image is 0.5 cm.
D) The size of the image is 0.4 cm.

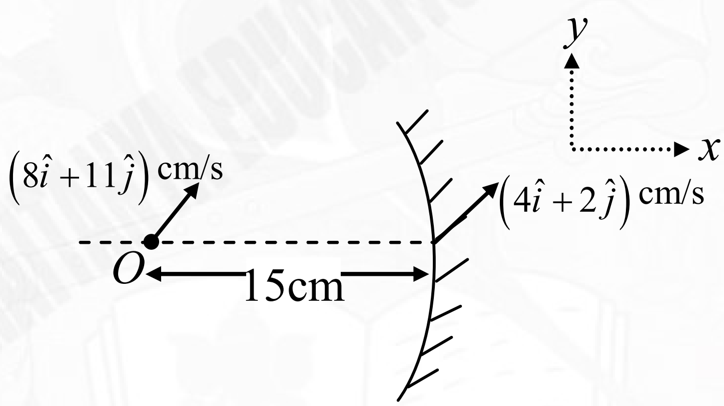
SECTION-IV
(INTEGER ANSWER TYPE)

- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
- *Full Marks* : *+4* If ONLY the correct integer is entered;
- *Zero Marks* : *0* In all other cases.

17. Water (refractive index$=\frac{4}{3}$) in a tank is 19 cm deep. Oil of refractive index $\frac{7}{4}$ lying on water makes a convex surface of radius of curvature 6 cm and acts as a thin lens. An object S is placed 24 cm above the water surface. Its image is formed at X cm above the bottom of the tank. Then X=______



18. A point object is located at a distance 15 cm from the pole of a concave mirror of focal length 10 cm on its principal axis is moving with a velocity $(8\hat{i}+11\hat{j})$ cm/s and velocity of mirror is $(4\hat{i}+2\hat{j})$ cm/s as shown. If $\vec{V}$ is the velocity of image. Then find the value of $\left|\frac{\vec{V}}{5}\right|$ (in cm/s) at the instant shown in figure



19. The principal section of glass prism is an isosceles $\triangle PQR$ with $PQ = PR$. The face PR is silvered. A ray is incident perpendicularly on face PQ and after two reflections, it emerges from base QR, normal to it. The angle of the prism is given by $\frac{\pi}{\alpha}$ rad. Find the value of α.

CHEMISTRY

Max. Marks: 60

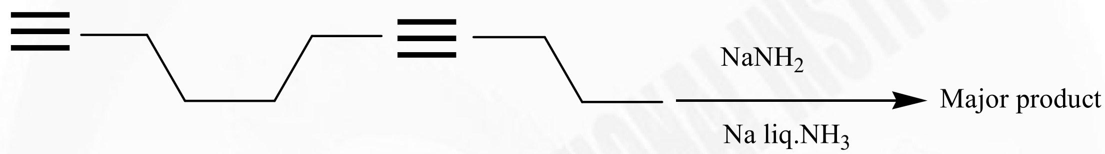
SECTION-I
(SINGLE CORRECT ANSWER TYPE)

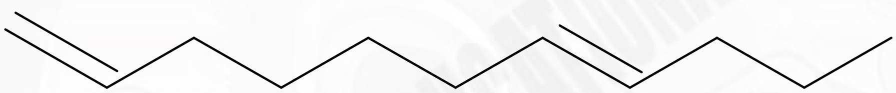
This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

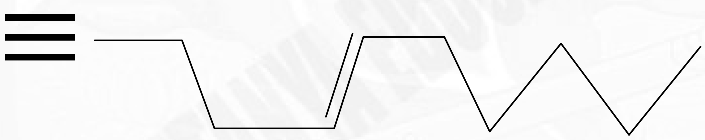
Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)

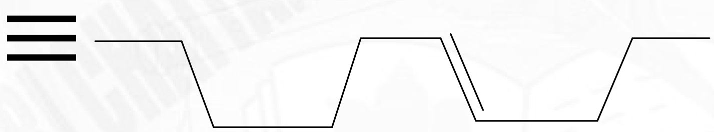
- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
- Full Marks: **+3** (If the answer is correct)
- Zero Marks: **0** (If the question is unanswered)
- Negative Marks: -1 (In all other cases.

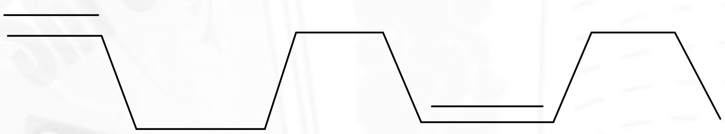
20.



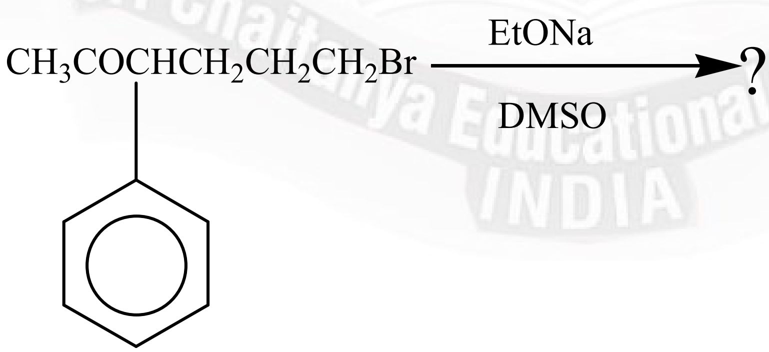
A) 

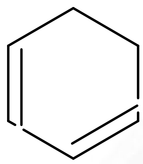
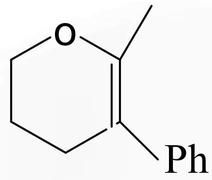
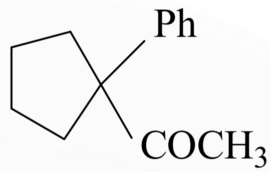
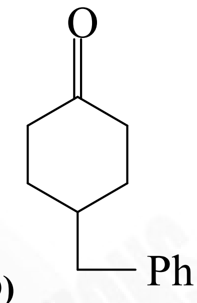
B) 

C) 

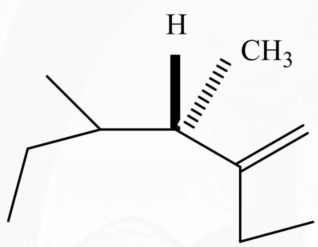
D) 

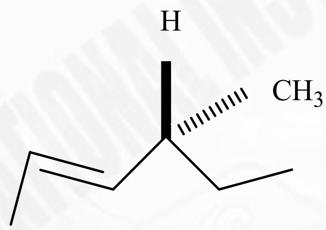
21. The major product formed in the following reaction is

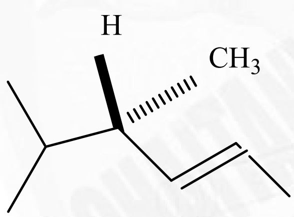


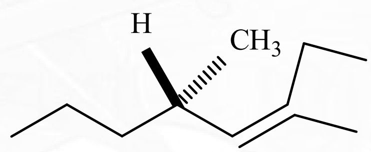
A)  B)  C)  D) 

22. Which of the following compounds produces optically inactive compound on catalytic hydrogenation?

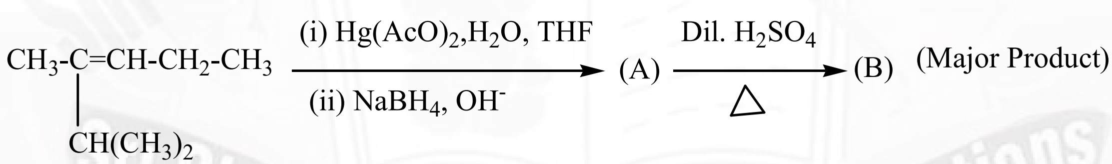
A) 

B) 

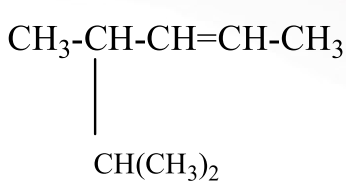
C) 

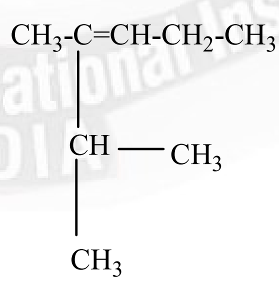
D) 

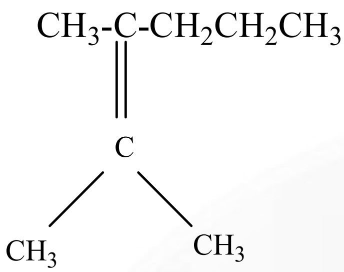
23.

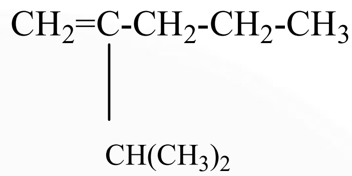


The major product (B) in the above sequence reaction is

A) 

B) 

C) 

D) 

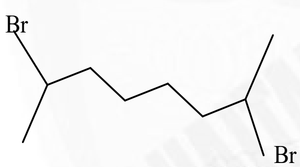
SECTION-II
(PARAGRAPH WITH NUMERICAL VALUE TYPE)

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** : **+2** If ONLY the correct numerical value is entered at the designated place;
- ***Zero Marks*** : ***0*** In all other cases.

Question Stem for Question Nos. 24 and 25

Question Stem

Consider the following reactions sequence



$$\xrightarrow[\text{dryether}]{Na} P \xrightarrow[1eq]{Br_2/h\gamma} Q \xrightarrow{alc.KOH} R \xrightarrow[Zn/H_2O]{o_3} S$$

$$R \xrightarrow[\text{Mono bromination}]{Br_2/hv} U_{(\text{isomeric products})} \xrightarrow[\Delta]{alc.KOH} V\ (\text{isomeric products})$$

24. How many deuterium atoms present in (Z)

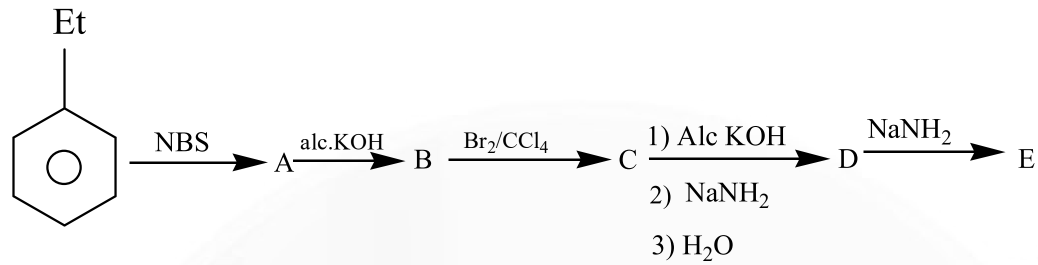
$$S \xrightarrow[\text{on long standing}]{D_2O/D^+} Z$$

25. Number of Isomeric products in (V) are ____

Question Stem for Question Nos. 26 and 27

Question Stem

A scholar worked out following reaction sequence



26. (How many primary Hydrogens present in the following major product (G)

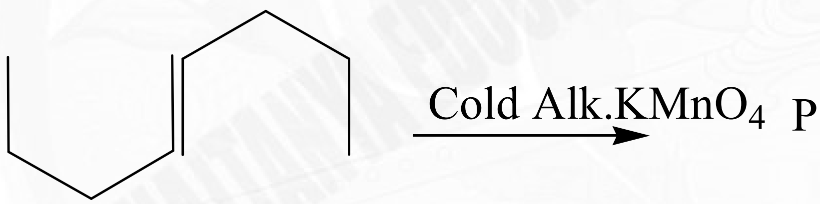
$$(E) \xrightarrow{CH_3I} F \xrightarrow{D_2/Pd\text{ - }C} G(\text{major})$$

27. Molecular weight of (I) is.

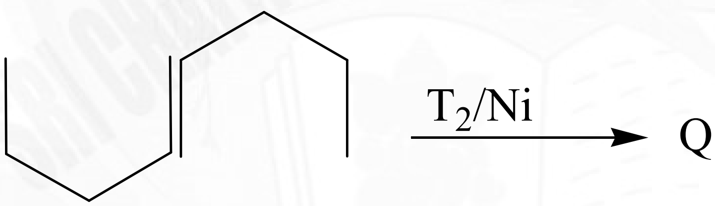
$$(B) \xrightarrow[2)\ NaBD_4,\ OD^-]{1)\ Hg(AcO)_2\ H_2O} (H) \xrightarrow[\Delta]{H+} (I)\text{ major}$$

Question Stem for Question Nos. 28 and 29

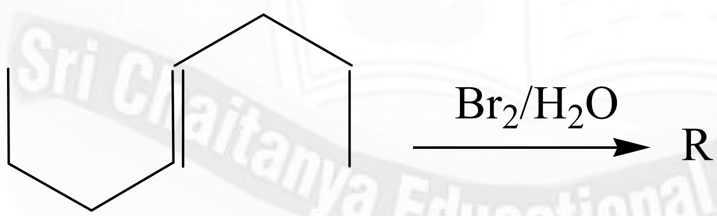
Question Stem



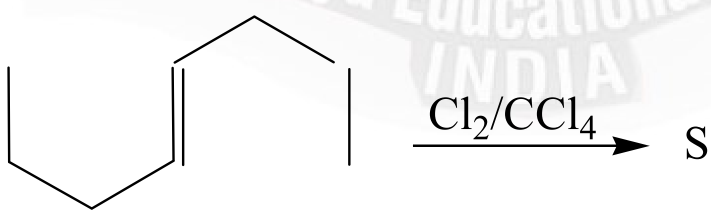
I)



II)



III)



IV)

28. How many are correct statements (if need will be there consider major products)

(i) 'P' is optically inactive due to internal compensation

(ii) 'Q' is optically active due to external compensation

(iii) 'R' is optically active due to internal compensation

(iv) 'S' is optically active due to external compensation

(v) Above all reactions are stereo specific reaction but not stereo selective

(vi) molecular weight of 'Q' is 114

29. Sum of products (P+Q+S) in I,II and IV reactions are __________

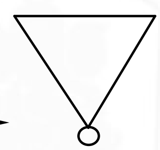
(if need will be there consider major products)

SECTION-III
(ONE OR MORE CORRECT ANSWER TYPE)

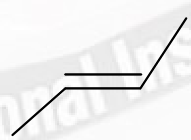
- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks: +3*** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks: +2*** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks: +1*** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks: 0*** If unanswered;
- ***Negative Marks: -2*** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

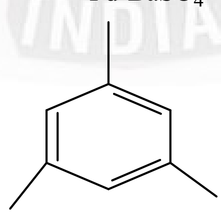
Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;
Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
Choosing any other option(s) will get -2 marks.

30. Select correct options

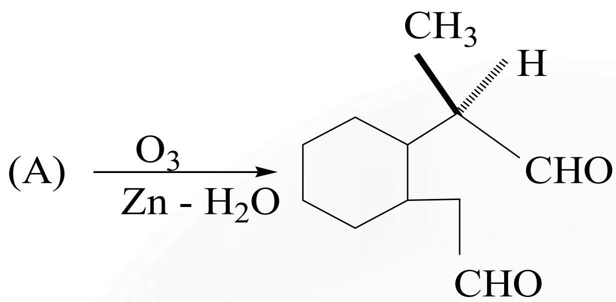
A) $CH_2{=}CH_2 + 1/2\ O_2 \xrightarrow{Ag}$ 

B) $CH_3\text{-}CH{=}CH_2 \xrightarrow[CCl_4]{NOCl} CH_3\text{-}CH(Cl)\text{-}CH_2\text{-}NO$

C) $CH_3\text{-}C{\equiv}CH \xrightarrow{KNH_2} \xrightarrow{CH_3Br} \xrightarrow[\text{Pd-BaSO}_4]{H_2}$ 

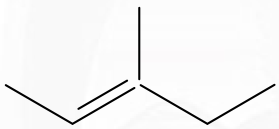
D) $CH_3\text{-}C \equiv CH \xrightarrow[\text{Fe tube}]{\text{Red hot}}$ 

31.

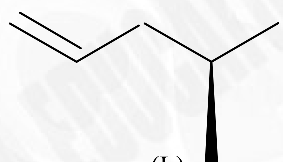


A) 'A' is Optically active.
B) 'A' reacts with Br_2 / CCl_4 racemic mixture obtained
C) 'A' reacts with Br_2 / CCl_4 diastereomeric mixture will be formed
D) 'A' reacts with O_3 / H_2O_2 dicarboxylic Acid formed

32. Choose the INCORRECT statement(s) about the following alkenes is/are



(K)

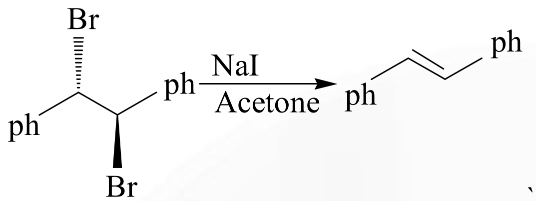


(L)

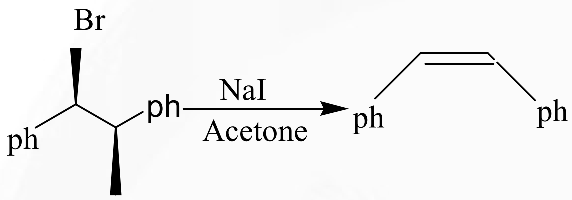
A) Alkene [K] is more stable than alkene [L].
B) Hydrogenation of achiral alkene [K]will give chiral product.
C) Hydrogenation of achiral alkene [L] will give chiral product.
D) The two alkenes [K] and [L] are structural isomers.

33. An alkane (A) having M.F C_6H_{14} generates 8 isomeric monochloroderivates and it can be prepared by cataltic hydrogenation of 5 optically inactive isomeric alkenes with M.F C_6H_{14} B is an isomer of A and it also generates 8 isomeric Monochloroderivates then correct statements about A & B is/ are?
A) B can be prepared by catalytic hydrogenation of 4 isomeric alkens.
B) Boiling point of A is more than B
C) A generate 6 enantiomers on mono chlorination while B also generate 3 enantiomeric pairs on mono chlorination
D) A prepared from 2 isomeric ketones while B also prepared from 2 isomeric ketones

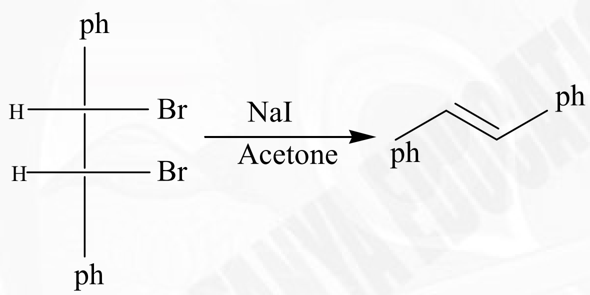
34. Identify the CORRECTLY matched reactant, reagent & product set (s)



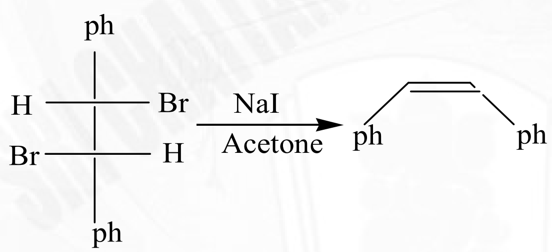
A)



B)

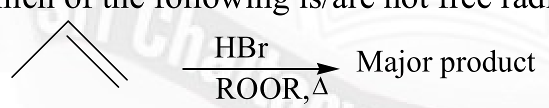


C)

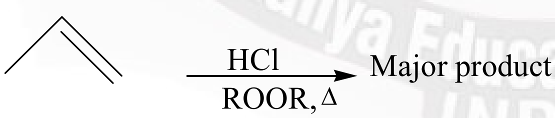


D)

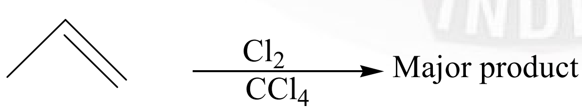
35. Which of the following is/are not free radical addition reactions



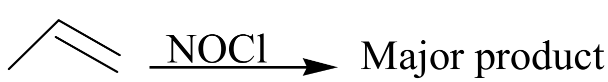
A)



B)



C)

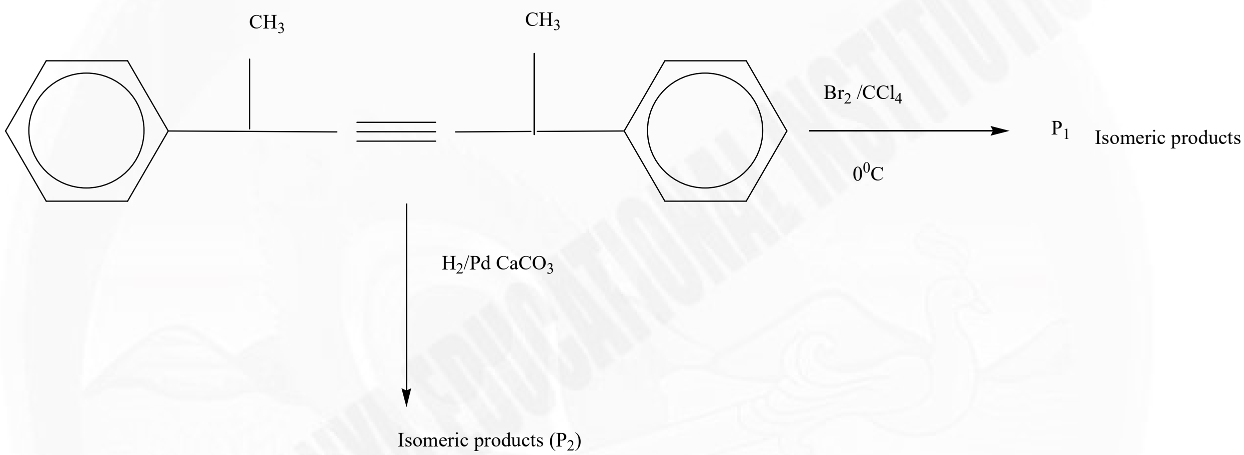


D)

SECTION-IV
(INTEGER ANSWER TYPE)

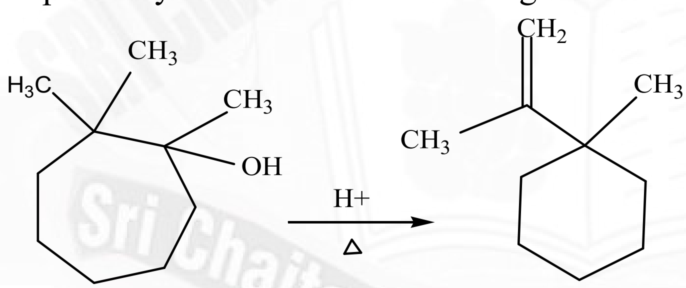
- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** ***: +4*** If ONLY the correct integer is entered;
- ***Zero Marks*** ***: 0*** In all other cases.

36.

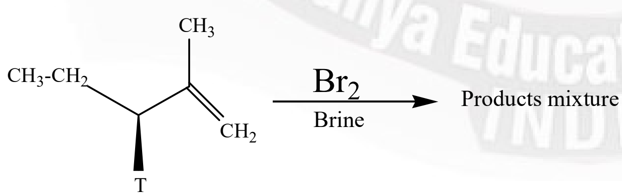


Sum of chiral products in P_1 and P_2 is

37. Number of transition states and number of reaction intermediates are X and Y respectively involved in the following reaction. sum of X+Y is.(excluding stereo isomers)



38.



Find the number of fractions obtained after fractional distillation of products mixture.(

Note: Neglect nucleophile attacking on primary carbon resulting product)

MATHEMATICS

Max. Marks: 60

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
- Full Marks: **+3** (If the answer is correct)
- Zero Marks: **0** (If the question is unanswered)

Negative Marks: -1 (In all other cases

39. Let $f(x)=\begin{cases} xe^{ax} & ;x\le 0 \\ x+ax^2-x^3 & ;x>0 \end{cases}$ where a is constant. In which of the following intervals $f^1(x)$ is increasing

A) $\left(-\infty,\frac{a}{3}\right),a<0$ **B)** $\left(-\infty,\frac{-2}{a}\right)\cup\left(\frac{a}{3},\infty\right)$

C) $\left(-\frac{2}{a},\frac{a}{3}\right),a>0$ **D)** $\left(-\frac{2}{a},\infty\right),a<0$

40. Let $f(x)=\begin{cases} \max.\left\{t^3-t^2+t+1.0\le t\le x\right\},0\le x\le 1 \\ \min.\left\{3-t,1<t\le x\right\} \quad 1<x\le 2 \end{cases}$

And $g(x)=\begin{cases} \max.\left\{\frac{3}{8}t^4+\frac{1}{2}t^3-\frac{3}{2}t^2+1,0\le t\le x\right\},\ 0\le x<1 \\ \min.\left\{\frac{3}{8}t+\frac{1}{32}\sin^2\pi t+\frac{5}{8},1\le t\le x\right\} \quad 1\le x\le 2 \end{cases}$

Which of the following options are does not holds good with respect to the given functions $f(x)$ and $g(x)$

A) $f(x)$ is continuous $\forall x\in[0,2]$ and not differentiable at atmost 2 points in $[0,2]$

B) $\lim\limits_{x\to 1^-} fog(x)>\lim\limits_{x\to 1^+} gof(x)$

C) Let $Z(x)=\frac{d}{dx}\left(f(x)^{g(x)}\right)$ *and* $Y(x)=\frac{d}{dx}\left(g(x)^{f(x)}\right)$ then $Z(x)$ and $Y(x)$ vanish simultaneously at no real value of x

D) Number of solutions of $f(x)=g(x)$ is 4

41. The complete set of non-zero values of k such that the equation $\left|x^2-10x+9\right|=kx$ is satisfied by atleast one and atmost three values of x, is

A) $(-\infty,-16]\cup[4,\infty)$ **B)** $(-\infty,-16]\cup[16,\infty)$

C) $(-\infty,-4]\cup[4,\infty)$ **D)** $(-\infty,4]\cup[16,\infty)$

42. If the coordinates of one of the two points on the curve $y=x^4-2x^2-x$ that have a common tangent line, can be (α,β), then the value of $\left\|\alpha\right|-\left|\beta\right\|=$

A) 1 **B)** 0 **C)** 8 **D)** 4

SECTION-II
(PARAGRAPH WITH NUMERICAL VALUE TYPE)

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +2*** If ONLY the correct numerical value is entered at the designated place;
- ***Zero Marks: 0*** In all other cases.

Question Stem for Question Nos. 43 and 44

Question Stem

Let $f(x)=\begin{vmatrix}1 & -1\\ -x & x^2-2\end{vmatrix}+\begin{vmatrix}-1 & 1\\ -2x & 2x^2-4\end{vmatrix}+\begin{vmatrix}1 & -1\\ -3x & 3x^2-6\end{vmatrix}+\begin{vmatrix}-1 & 1\\ -4x & 4x^2-8\end{vmatrix}+.......+\begin{vmatrix}1 & -1\\ -31x & 31x^2-62\end{vmatrix}$

If P: represent minimum value of $g(x)$ where $g(x)=\int_0^x f(x-t)dt, x\in[0,3]$

And Q: represent the value of (a+b), if the set of values of k for which the equation $\left|f(x^2)\right|=k$ has six different solutions is (a,b), then

43. The number of positive integral divisors of Q-3P=

44. The value of the remainder when $(Q-4)^{-P}$ is divided by 100 is

Question Stem for Question Nos. 45 and 46

Question Stem

Let $f(x)=x^3-(6\sin\theta)x^2+(2\sin^2\theta+q)x-1,\ q,\theta\in R$ be a cubic polynomial in x. If exactly one tangent can be drawn to the graph of $y=f(x)$ which is parallel to the line y=x, then

45. The largest value of q is

46. If q is smallest then the value of $13.\frac{d}{dx}\left(f^{-1}(x)\right)\Big|_{x=9}=$

Question Stem for Question Nos. 47 and 48

Question Stem

If a differentiable function 'f' satisfies the relation

$\int_0^{t^2} f^2(x)dx+\frac{1}{2}\int_0^{\pi/2} t^6\sin^3 x\,dx=\int_0^{t^2} 2xf(x)dx\ \forall t\in R$, then find the number of solution (s) of the equation

47. $f(x)=\sqrt[3]{x}$

48. $f(x)=-\sin^{-1}(\sin x)$

Where inverse trigonometric function takes principal values

SECTION-III

(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** : **+4** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks*** : **+3** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks*** : **+2** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks*** : **+1** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks*** : **0** If unanswered;
- ***Negative Marks: -2*** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;
Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
Choosing any other option(s) will get -2 marks.

49. Let S be the set of all twice differentiable functions f from R to R such that $f^{11}(x) < 0 \forall x \in (-2,2)$ for $f \in S$, let X_f be the number of points $x \in (-1,1)$ for which $f(2x) = 3x$. Then which of the following options is/are **correct**?

A) There exists a function $f \in S$ such that $X_f = 0$.

B) For every function $f \in S$, we have $X_f \leq 2$.

C) there exists a function $f \in S$, we have $X_f > 2$.

D) There does NOT exist any function $f \in S$ such that $X_f = 1$.

50. Let $\alpha = \sum_{k=1}^{\infty} \sin^{2k}\left(\frac{\pi}{6}\right)$

Let $g:[0,1] \to R$ be the function defined by $g(x) = 2^{\alpha x} + 2^{\alpha(1-x)}$

Then, which of the following statements is/are TRUE?

A) The minimum value of $g(x) is 2^{7/6}$

B) The maximum value of $g(x)$ $is 1 + 2^{1/3}$

C) The function $g(x)$ attains its maximum at more than one point

D) The function $g(x)$ attains its minimum at more than one point

51. Let $f(x) = \frac{(x^3 - 3x + 1)(x - \alpha)}{(x - \alpha)}$

If range f(x) is a proper subset of real numbers, then Which of the following is/are correct?

A) range of α can be $(-\infty, -2) \cup (2, \infty)$

B) number of integral values that α cannot take is 5

C) $\lim_{x \to \alpha} f(x) \in A \text{ then set } A \text{ can be} (-\infty, -1) \cup (3, \infty)$

D) The least positive integral value of α is 3

52. Which of the following statement is (are) correct?

A) Let $f:[0,1]\to R$ be a differentiable function with $f(0)=0,$ and $0\le f^1(x)\le 2f(x),$ then $f(x)$ is identically zero $\forall X\in[0,1].$

B) If $f(x)=x^3+ax^2+bx+5\sin^2 x$ is an increasing function $\forall x\in R,$ then $a^2-3b+15\le 0.$

C) If $x^a\le a^x$ $\forall x\in(1,\infty)$ where $a\in(1,\infty)$ then number of value of a satisfying the inequality is exactly one.

D) Let $f(x)$ be a differentiable function in $[0,\infty)$ with $f(0)=0$ and $f^1(x)$ is increasing function $\forall x\geq 0,$ then $g(x)=\begin{cases}\dfrac{f(x)}{x}, & x>0\\ f^1(0), & x=0\end{cases}$ is a decreasing function.

53. Let $f:R\to R$ be a continuous and differentiable function such that $f(2)=3,$ $f(6)=8$ and has local minimum at $x=2.$ If $g(x)=\int_2^6 f(2x+t)dt,$ then

A) $g^1(0)=5$

B) $g^1(0)=10$

C) $\lim_{x\to 0}\dfrac{g\left(f\left(x+2\right)-3\right)-g(0)}{x}=0$

D) $\lim_{x\to 1}\dfrac{f(6x)-f(6)}{x-1}=\lim_{x\to 0}\dfrac{6f(x+6)-16f(x+2)}{x}$

54. Consider a polynomial function $f(x)=x^5+ax^4+bx^3+cx^2+dx+e.$ if the line $y=2$ touches it at $x=1$ and $x=4$ where $x=4$ is also a point of inflection of $y=f(x),$ then

A) at $x=\dfrac{11}{5,}f(x)$ has a local minimum.

B) at $x=\dfrac{11}{5,}f(x)$ has a local maximum.

C) the value of $f(2)+f(3)$ equal-8.

D) the value of $\int_2^6 f(x)dx\ is\dfrac{424}{5}.$

SECTION-IV
(INTEGER ANSWER TYPE)

- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** : ***+4*** If ONLY the correct integer is entered;
- ***Zero Marks*** : ***0*** In all other cases.

55. If $f(x)=x^3-3x+\sin^{-1}(a^2-3a+2)$. Then the smallest positive integer 'a' for which $f(x)=0$ has three distinct real solution.

56. Let the function $f:R\to R$ be defined by

$$f(x)=\frac{\sin x}{e^{2024x}}\frac{\left(x^{2025}+2024x+2025\right)}{\left(x^2-x+1\right)}+\frac{2}{e^{2024x}}\frac{\left(x^{2025}+2024x+2025\right)}{\left(x^2-x+1\right)}$$

Then the number of solutions of $f(x)=0$ in R is ________

57. Consider the functions $f,g:R\to R$ defined by $f(x)=x^2+\frac{5}{12}$ and

$$g(x)=\begin{cases}2\left(1-\frac{4|x|}{3}\right), & |x|\le\frac{3}{4}\\ 0, & |x|>\frac{3}{4}\end{cases}$$, and $h(x)=\{\min\{f(x),g(x)\}\}$, then the number of points of local maximum or local minimum of $h(x)$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KALRA RISHI HT.No. 237047209
3 RANK
9th - 12th Class Sri Chaitanya
RAGHAV GOYAL HT.NO. 237020329
4 RANK
4th - 12th Class Sri Chaitanya
BIKKINA ABHINAV CHOWDARY HT.NO. 236163128
7 RANK
32 TOP RANKS BELOW 100 IN ALL-INDIA OPEN CATEGORY
RANK 12 DASARI SAKETH NAIDU HT.No. 236060148
RANK 14 GUTHIKONDA ABHIRAM HT.No. 236129121
RANK 17 P GNANA KOUSIK REDDY HT.No. 236133520
RANK 18 D VENKATA YUGESH HT.No. 236049359
RANK 19 MOULIK JINDAL HT.No. 237021004
RANK 22 D PRATAP SINGH HT.No. 232006139 (DLP)
RANK 23 SHAIK SAHIL CHANDA HT.No. 236132870
RANK 26 SONI MAYANK HT.No. 232097025 (DLP)
RANK 33 GOYAL NAMAN HT.No. 232029345 (DLP)
RANK 37 SUMEDH SS HT.No. 231029088 (DLP)
RANK 38 V DEERAJ SATHVIK REDDY HT.No. 236057151
RANK 44 V VERGIYA KAUSHAL HT.No. 231004058 (DLP)
RANK 46 SUTHAR HARSHAL HT.No. 231005060 (DLP)
RANK 58 ABHI JAIN HT.No. 232091036 (DLP)
RANK 61 RIDAM JAIN HT.No. 232068021 (DLP)
RANK 62 SUTHAR HARSH HT.No. 232100028 (DLP)
RANK 63 SOJITRA DIPEN HT.No. 231017001 (DLP)
RANK 65 D PANEENDRANADHA REDDY HT.No. 236050298
RANK 67 GUNDA SUSHANTH HT.No. 236129732
RANK 68 HARSH TAYA HT.No. 237021178
RANK 69 KRISH GUPTA HT.No. 232068189 (DLP)
RANK 73 KUNCHAKURTHY VIVEK HT.No. 236134093
RANK 81 N DHARMA TEJA REDDY HT.No. 236050138
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